

- 2 Blood transports many substances through the body, including gases such as oxygen and carbon dioxide.

(a) Water is the main component of blood.

Explain how the properties of water allow the blood circulatory system of mammals to function.

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..... [3]

(b) State **two** different ways in which carbon dioxide is transported in the blood.

1

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2

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[2]

[illegible]

Figure 2.1 is a ribbon diagram showing the structure of myoglobin.

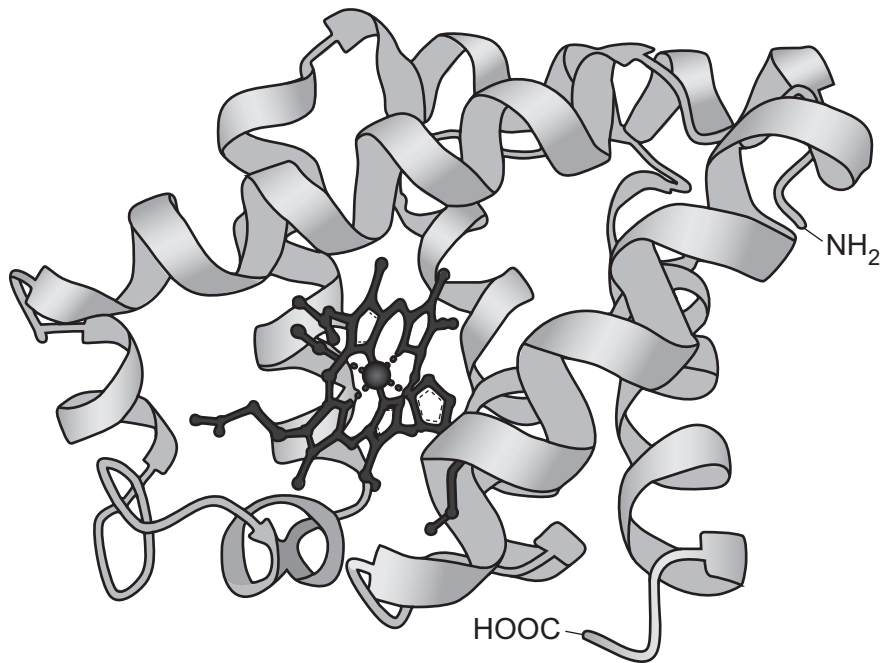


Figure 2.1

- (i) Describe similarities **and** differences between the structure of the myoglobin molecule shown in Figure 2.1 and the structure of a haemoglobin molecule.

[4]

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

- (ii) Contracting muscles have a high requirement for oxygen, which is supplied to muscle cells by the haemoglobin in red blood cells.

Myoglobin has a higher affinity for oxygen than haemoglobin.

Suggest why it is important for muscle function that myoglobin has a higher affinity for oxygen than haemoglobin.

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..... [2]

[Total: 11]